Anticipatory Behavioural Responses to Environmental Policy Announcements: Evidence from the London Ultra Low Emission Zone

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Abstract

ABSTRACT

1 Introduction

Standard models of regulatory compliance assume behavioural change occurs at the point of implementation. Yet if economic agents are forward-looking and form rational expectations, as Muth (1961) first formalised, credible future policy may alter present behaviour well before implementation begins. Kydland and Prescott (1977) demonstrated that the effectiveness of policy depends fundamentally on its credibility: agents who believe a policy will be implemented adjust accordingly, while announcements perceived as reversible or discretionary will be ignored. Lucas (1976) showed more generally that behavioural models are not invariant to policy regime, since the decision rules being estimated already embed agents' expectations about future policy. Applied to the gap between announcement and enforcement, this implies that treating behaviour as static until formal implementation misspecifies the underlying decision problem. Credible policy announcements may therefore constitute a policy instrument in their own right.

The London Ultra Low Emission Zone (ULEZ) provides a natural empirical setting for studying anticipatory compliance. Introduced in April 2019 under Mayor Sadiq Khan, the scheme charges non-compliant vehicles a daily fee for entering the zone, with the stated aim of reducing the air pollution damaging residents' health and the environment. Two features make it a useful case. First, each expansion was announced well before it took effect. The London-wide expansion of August 2023, for instance, was proposed in March 2022 and confirmed that November, leaving a gap of roughly nine months between the November confirmation and enforcement. That gap provides a window in which anticipatory compliance, if it occurs, can be detected. Second, the zone expanded three times between 2019 and 2023, each announced in advance of implementation. Repeated expansions allow the anticipatory response to be tested across successive waves, and they raise a further question: whether the effect weakens as each new announcement conveys less genuinely new information than the last.

The paper addresses two questions. First, did each ULEZ expansion generate anticipatory compliance before implementation? Second, does the anticipatory effect vary across the three expansions? Both follow from the theoretical claim at the centre of this study: that a credible announcement functions as a policy instrument in its own right, capable of shifting behaviour before enforcement. If this holds, compliance should rise measurably in the window between a credible announcement and its implementation. Whether it does, and whether the response weakens as successive expansions convey less new information, is what the analysis sets out to test.

The analysis uses synthetic control as its primary method. For each treated borough, the method builds a counterfactual from a weighted combination of untreated units, which avoids reliance on the parallel trends assumption required by conventional difference-in-differences. The donor pool for each expansion draws on units not yet treated in that wave: for the 2019 and 2021 expansions this includes outer London boroughs alongside the surrounding districts, while for the 2023 London-wide expansion, where every London borough is eventually treated, the donor pool rests on the surrounding never-treated districts alone. The dataset covers 52 units: 33 London boroughs and 19 surrounding districts serving as never-treated controls.

These form a balanced quarterly panel running from 2011 Q4 to 2024 Q4. The outcome is the ultra low emission vehicle (ULEV) share of licensed cars, constructed from DVLA vehicle licensing records. The synthetic control results are tested against two-way fixed effects difference-in-differences, an event study, and interrupted time series specifications, with a geographic regression discontinuity design considered but limited by the small number of boroughs near the zone boundary.

The analysis finds anticipatory compliance in specific boroughs rather than spread evenly across London, and the effects are consistently modest. For the 2019 expansion, Tower Hamlets shows the strongest evidence of a response, with its ULEV share rising above its synthetic counterfactual in the quarters after the announcement, though the effect is marginal rather than conventionally significant. The 2021 expansion yields no significant effects, with only Hammersmith and Fulham reaching the suggestive range. For the 2023 London-wide expansion, two outer boroughs show significant anticipatory effects once surrounding never-treated districts are used to build the counterfactual: Hillingdon and Sutton.

This study makes three contributions. Theoretically, it extends the literature on anticipatory behaviour, developed largely outside environmental regulation, to a domain where compliance requires a costly and durable purchase. Empirically, it tests this mechanism across three repeated waves of the same policy, where existing work has largely examined single policy events, tracing the anticipatory response as the policy moves from novel to familiar. Methodologically, it uses districts bordering Greater London as never-treated controls, a donor pool that shares the commuter exposure of outer London boroughs without being subject to the zone itself, providing a cleaner counterfactual than inner London boroughs that were themselves treated in earlier waves.

ROADMAP

2 Policy Background

The Ultra Low Emission Zone (ULEZ) is a charging scheme run by Transport for London (TfL) under the Mayor of London. Vehicles that do not meet specified exhaust emission standards pay a daily fee of £12.50 to drive within the zone, applied at all hours. Petrol cars and vans must meet the Euro 4 standard, diesel cars and vans the Euro 6 standard, and motorcycles and mopeds the Euro 3 standard. Compliance is defined by these emission standards rather than by fuel type: a Euro 4 petrol or Euro 6 diesel car is exempt from the charge, as are fully electric and most plug-in hybrid vehicles. Ultra low emission vehicle adoption is therefore one route to compliance among several, but a distinctive one, since it represents the durable, forward-looking capital decision that anticipatory behaviour would predict. The policy aims to improve air quality by lowering concentrations of harmful pollutants, particularly nitrogen dioxide and particulate matter, and to protect public health across London.

ULEZ emerged as a gradual evolution of London's clean air policy, building on the 2003 Congestion Charge and the 2008 Low Emission Zone that targeted heavy vehicles. Introduced to address illegal levels of air pollution, the scheme was accelerated and launched in central London by Mayor Sadiq Khan in 2019, driven by rising public health concerns and a series of legal defeats over the UK's failure to meet air quality obligations. These earlier schemes meant London drivers were already familiar with zone-based charging, and the enforcement infrastructure of cameras and Automatic Number Plate Recognition (ANPR) was already in place. That existing apparatus made subsequent ULEZ announcements credible: a commitment to expand the zone was believable precisely because the equipment to enforce it already existed.

Launched on 8 April 2019, London's initial 24-hour ULEZ was mapped to the central Congestion Charge Zone, covering roughly 21 square kilometres and encompassing the high-density commercial core. The scheme was first proposed in 2015 under Mayor Boris Johnson, but the decisive signal came on 4 April 2017, when Mayor Sadiq Khan announced the April 2019 launch date and brought the timetable forward. The same 2017 announcement also flagged a planned extension to the North and South Circular roads, which was formally confirmed on 8 June 2018 with an implementation date of 25 October 2021. This April 2017 announcement is the point from which anticipatory responses to the central zone are measured in the analysis that follows.

The inner expansion to the North and South Circular roads was confirmed in June 2018 and implemented on 25 October 2021. Charges and emission standards remained identical to the central zone, but the scale increased sharply, covering an area roughly eighteen times larger and bringing approximately 3.8 million residents within the zone [Mayor of London Press release]. The gap of over three years between confirmation and enforcement is the longest of the three expansions, giving the widest window in which anticipatory compliance could occur.

The scheme was then proposed for a London-wide expansion in March 2022, confirmed in November 2022, and implemented on 29 August 2023, leaving a nine-month window between confirmation and enforcement. This was the largest expansion, bringing all 32 boroughs and the City of London within the zone (Figure 1). Its credibility was tested directly: a coalition of five councils — the outer London boroughs of Bexley, Bromley, Harrow and Hillingdon, together with Surrey County Council — challenged the expansion in the High Court, but the challenge failed in July 2023 and enforcement proceeded as announced. That the commitment held despite legal challenge reinforced the signal to drivers that the deadline was firm.

The credibility of these commitments rested on three reinforcing foundations. Each expansion was preceded by statutory public consultation and formal confirmation, signalling that the policy followed a legally prescribed procedure rather than an arbitrary mayoral decision. The commitment to charge non-compliant vehicles was underpinned by the camera and ANPR infrastructure already in place, so the announced restrictions could be monitored and penalties imposed from the first day of enforcement. Most decisive was demonstrated follow-through: the 2019 and 2021 expansions were both delivered on their announced timetables, so by the time the 2023 expansion was confirmed, drivers had twice seen the Mayor commit to a future extension and carry it out. Announcements that might once have been read as provisional intentions had become commitments road users could expect to be honoured. Credible, pre-announced expansions created precisely the conditions under which forward-looking individuals would adjust their behaviour before enforcement began.

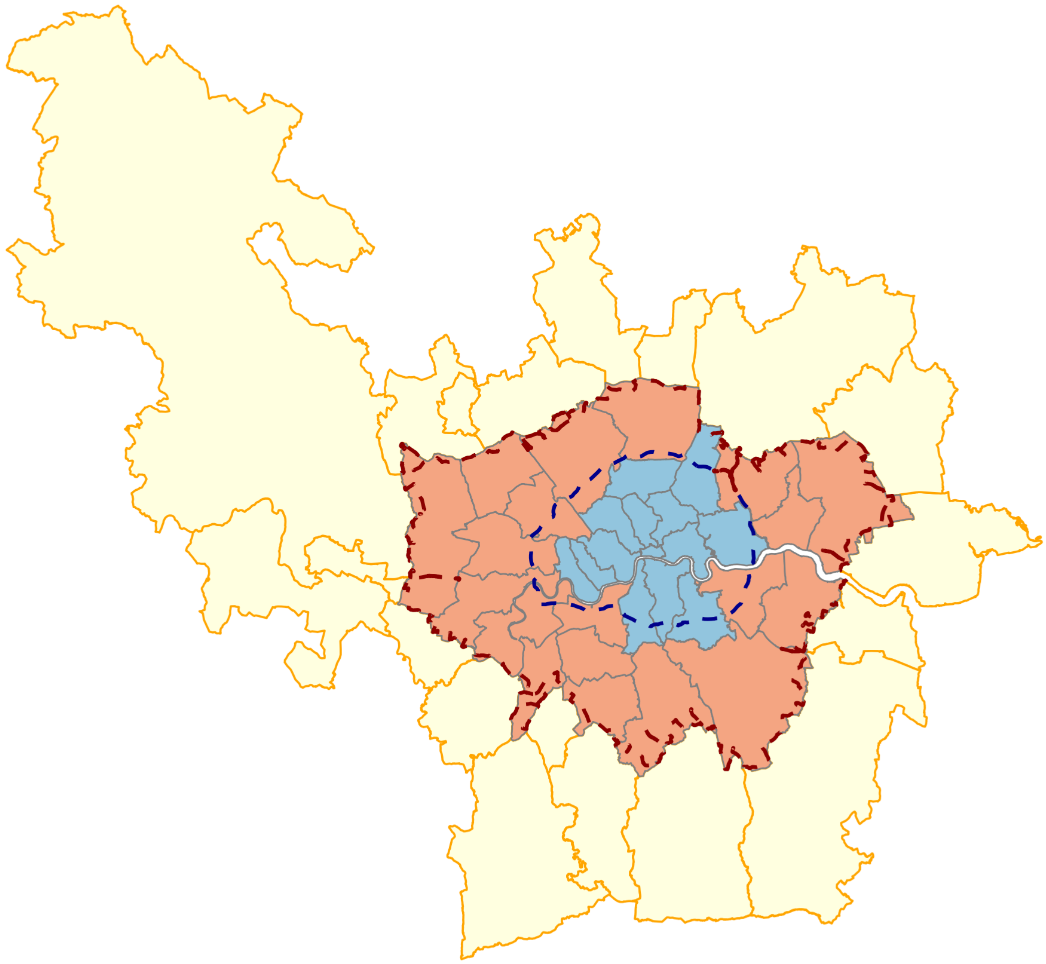

Figure 1: Study area and treatment assignment. Inner London boroughs (blue) were covered by the 2019 and 2021 expansions; outer London boroughs (red) by the 2023 London-wide expansion; the nineteen surrounding districts (yellow) were never subject to the charge. Dashed lines mark the 2021 and 2023 zone boundaries.

3 Literature Review

This literature review establishes the foundations for the dissertation in three stages. It first develops the theoretical framework, which explains why credible policy announcements should induce anticipatory behavioural responses before implementation. It then reviews the existing empirical evidence on anticipatory behaviour and low-emission zone policies, assessing what is already known and where important questions remain unanswered. Finally, it turns to the methodological literature, outlining the empirical strategies capable of identifying such anticipation and motivating the research design adopted here. The review concludes by setting out how the research design responds to that gap.

3.1 Theoretical Framework

The theoretical foundation for anticipatory behavioural responses begins with the premise that economic agents make decisions using information about both current and expected future conditions. John F. Muth's rational expectations hypothesis provides the micro-foundation for this argument. Muth (1961) proposed that agents' expectations of future economic variables are not systematically biased but instead coincide, on average, with the predictions implied by the relevant economic model, conditional on the information available to them. Expectations formed this way efficiently incorporate all available information, including credible signals about future policy, rather than relying solely on past experience or adjusting only once change occurs. Although Muth developed this framework in the context of commodity price expectations, its logic extends naturally to announced regulatory costs. When policymakers communicate a future increase in the cost of owning or operating a non-compliant vehicle, that information becomes relevant immediately rather than only when the regulation enters into force. A forward-looking vehicle owner therefore has an incentive to evaluate the future costs of retaining a non-compliant vehicle against the costs of replacing or scrapping it before enforcement begins. Under rational expectations, disregarding information already revealed would itself be irrational, even though the timing of any individual response depends on the costs of adjustment.

This forward-looking behaviour has important implications for how policy effects should be understood empirically. If agents incorporate anticipated policy into present decisions, then observed behaviour before implementation may already reflect the future policy environment. Lucas (1976) formalised this insight through the Lucas critique, arguing that the estimated relationships used to evaluate policy cannot be assumed to remain invariant across policy regimes, because those relationships already embed agents' expectations about the policy environment. Once a future policy is announced, agents revise their decision rules to reflect the expected future environment. Consequently, the period preceding implementation is not necessarily an undisturbed baseline against which policy effects can simply be measured. While the Lucas critique is often invoked as a warning against evaluating policy using naive before-and-after comparisons, it is more constructive in the present context. The possibility that behaviour changes before enforcement is not a methodological nuisance to be eliminated but the phenomenon of substantive interest. Rather than treating pre-implementation adjustment as a source of bias, this paper asks whether the announcement induced measurable changes in vehicle compliance before the ULEZ expansion came into force.

Whether such anticipatory adjustment occurs, however, depends on a further condition. Rational agents will respond to announced future policy only if they believe that the policy will in fact be implemented as announced. Kydland and Prescott (1977) demonstrate that expectations depend not only on policymakers' announcements but also on the credibility of those announcements. If agents expect a policy to be delayed, reversed, or weakened before implementation, the future regulatory cost remains uncertain and there is little reason to incur the costs of adjustment in advance. Conversely, where a policy commitment is regarded as credible and durable, the expected future cost becomes sufficiently certain to influence present behaviour. Credibility therefore provides the mechanism linking announcements to anticipatory responses. This insight is particularly important for understanding differences across successive ULEZ expansions. Variation in anticipatory compliance across waves need not imply that rational expectations fail; rather, it may reflect differences in the informational content or perceived credibility of successive announcements. As credible commitments accumulate over time, later announcements may convey less genuinely new information, reducing the scope of additional anticipatory adjustment even when the policy itself remains fully credible.

Taken together, these three contributions generate a coherent theoretical framework for analysing anticipatory behavioural responses. Rational agents incorporate credible information about future policy into current decision-making; those expectations imply that behavioural adjustment may begin before implementation, meaning that the pre-implementation period cannot automatically be treated as an unaffected baseline; and such adjustment occurs only when policy announcements are regarded as credible. The resulting prediction is therefore conditional rather than universal: where a credible announcement conveys precise information about a future regulatory cost, forward-looking agents have an incentive to adjust before implementation when the expected gains from acting early outweigh the costs of adjustment. Anticipatory responses should therefore be expected to be partial and heterogeneous rather than uniform across all individuals or policy settings. Although the components of this theoretical chain have been examined separately across a range of economic contexts, they have not been brought together to study anticipatory responses to geographically bounded environmental regulation in the window before implementation — the intersection this dissertation addresses.

3.2 Empirical Evidence

The theoretical framework established the conditions under which anticipatory responses should occur. The next question is whether such responses have been observed empirically. Existing evidence demonstrates that credible policy announcements can induce behavioural adjustment before implementation across a range of settings, while a separate literature shows that low-emission zone policies alter vehicle compliance. These two strands, however, have largely developed independently.

The closest empirical analogue is provided by Mills and Rüttenauer (2022), who examine behavioural responses to the announcement of mandatory COVID-19 certification. Using a synthetic control design, they show that vaccination rates begin increasing approximately twenty days before certification requirements entered into force, demonstrating that a credible announcement of a future access restriction altered behaviour before legal enforcement commenced. Their analysis explicitly backdates the intervention period to capture anticipation rather than treating implementation as the sole point of policy exposure, making it methodologically as well as conceptually similar to this paper. The principal distinction lies in the nature of the treatment. Whereas COVID certification constituted a national policy applying uniformly across the population, the ULEZ expansion is geographically bounded, leaving areas outside the zone untreated while subject to the same regional conditions. This clear separation between treated and untreated areas provides a sharper counterfactual than is available in a nationwide intervention. More broadly, Mills and Rüttenauer establish that anticipatory responses are observable in the context of public health policy, demonstrating the phenomenon outside environmental regulation while leaving open the question of whether similar behaviour arises in geographically targeted environmental policies.

Malani and Reif (2015) provide the conceptual framework for interpreting such anticipatory responses. Studying tort reform across US states, they argue that apparent pre-treatment trends in difference-in-differences designs need not indicate a violation of the parallel trends assumption but may instead reflect genuine behavioural responses to announced policy changes. This distinction is particularly important for this paper, where an increase in ULEZ-compliant vehicles before implementation could be interpreted either as an underlying trend or as anticipation. Their framework supplies the econometric language for adjudicating between these competing explanations, rather than presuming that pre-implementation movement is necessarily confounding. Their contribution also foreshadows the methodological discussion that follows by illustrating how empirical designs must explicitly account for anticipation when estimating policy effects.

Complementary evidence comes from Ito and Sallee (2018), who show that Japanese vehicle manufacturers strategically adjusted vehicle weight in anticipation of fuel-economy regulatory thresholds, building on the bunching framework developed by Saez (2010). Although these studies examine supply-side responses to regulatory notches rather than household vehicle replacement decisions, they reinforce the broader principle that known future policy thresholds induce strategic adjustment before they become binding. The underlying logic is therefore shared, even though the behavioural margin and institutional setting differ substantially from geographically bounded emission-zone policy.

A separate body of literature examines how low-emission zone policies influence vehicle fleet composition after implementation. Ellison, Greaves, and Hensher (2013) provide the foundational analysis of London's 2008 Low Emission Zone, documenting changes in heavy-vehicle fleet composition alongside modest reductions in particulate emissions using before-and-after comparisons. While their analysis is descriptive and does not employ a formal counterfactual, it establishes that DVLA vehicle licensing data capture meaningful fleet responses to emission-zone policy, providing empirical support for the outcome measure used in this dissertation.

The study most closely aligned with the present setting is Wolff (2014), who investigates Germany's Low Emission Zone programme and finds that vehicle owners living closer to an LEZ boundary were more likely to register cleaner vehicles, revealing a spatial gradient in compliance driven by proximity to the regulated area. This demonstrates that geographically defined environmental policies can generate measurable changes in vehicle registration behaviour. However, Wolff analyses compliance after the policy is already in force, whereas the present dissertation focuses on the anticipatory period between announcement and implementation. The distinction is important: existing evidence establishes that geography shapes compliance once regulation exists, but not whether geographically bounded announcements induce behavioural adjustment before enforcement begins.

Zhai and Wolff (2021) strengthen this literature through an instrumental variables analysis of London's Low Emission Zone, showing that compliance responses differed across implementation phases, with short-run adjustments diverging from longer-run environmental improvements. Their findings demonstrate that fleet responses evolve over time rather than following a simple monotonic path, providing important context for interpreting behavioural adjustment around successive policy expansions.

Taken together, these empirical literatures establish two complementary findings. First, credible policy announcements can induce anticipatory behavioural responses across a variety of policy domains. Second, geographically defined emission-zone policies produce measurable changes in vehicle compliance once implemented. However, no existing study brings these strands together by examining whether announcements of geographically bounded emission-zone expansions generate measurable compliance responses before implementation. Addressing that question requires an empirical strategy capable of identifying anticipation while constructing a credible counterfactual, to which the methodological literature reviewed below now turns.

3.3 Methodological Literature

The preceding section implied that identifying anticipatory behaviour requires an empirical design capable of satisfying two conditions: constructing a credible counterfactual without assuming an undisturbed pre-treatment period, and ensuring that comparison units remain genuinely unaffected by the policy announcement. Because policy anticipation may emerge before implementation, the appropriate methodological question is not simply whether outcomes change after enforcement, but which empirical strategies can credibly isolate behavioural responses to the announcement itself while providing valid inference with a small number of treated units.

The synthetic control method provides the principal framework for addressing these requirements. Introduced by Abadie and Gardeazabal (2003) and subsequently formalised by Abadie, Diamond, and Hainmueller (2010), the approach estimates the counterfactual outcome for a treated unit by constructing a weighted combination of untreated units that closely reproduces its pre-treatment trajectory. Rather than assuming that treated and untreated units would have followed parallel trends, the synthetic control estimator uses observed pre-treatment outcomes and predictors to approximate what would have occurred in the absence of treatment. This feature is particularly well suited to the present setting. As the theoretical discussion of anticipatory behaviour suggests, policy announcements may already influence behaviour before implementation, making the pre-treatment period an unreliable benchmark if treated and untreated units are simply assumed to evolve in parallel. Synthetic control instead constructs that benchmark directly from the data. Abadie, Diamond, and Hainmueller (2015) further develop placebo-based inference, assessing whether the estimated treatment effect is unusually large relative to placebo effects generated by assigning treatment to each donor unit in turn. Statistical significance therefore derives from the treated unit's position within the placebo distribution rather than from conventional sampling theory. Inference consequently depends on the size of the available donor pool, and placebo diagnostics remain informative alongside reported p-values. Abadie (2021) synthesises this literature and emphasises a further requirement that is central to the present dissertation: donor units must be plausibly unaffected by the intervention itself, while excessively large or heterogeneous donor pools may weaken the credibility of the constructed counterfactual.

The choice of comparison units is further informed by the literature on staggered policy adoption. Goodman-Bacon (2021) demonstrates that two-way fixed effects estimators in staggered treatment settings decompose into multiple two-by-two comparisons, some of which use already-treated units as controls for later-treated units. When treatment effects are heterogeneous or evolve over time, these comparisons can generate biased estimates and even negative weighting. Callaway and Sant'Anna (2021) address this problem by estimating group-time average treatment effects using only never-treated or not-yet-treated units as comparison groups. This framework is directly relevant to the successive ULEZ expansions, where treatment is introduced in multiple waves rather than simultaneously. It provides the methodological justification for constructing donor pools from units that have not yet been exposed to the relevant announcement and explains why conventional two-way fixed effects difference-in-differences serves as a robustness check rather than the primary empirical strategy, alongside complementary event-study and interrupted time-series specifications.

A complementary methodological perspective is provided by the geographic regression discontinuity literature. Keele and Titiunik (2015) establish the conditions under which administrative boundaries can identify causal effects, requiring continuity of potential outcomes at the boundary, the absence of coincident treatments, and sufficient observations close to the discontinuity. Dell's (2010) study of Peru's historical mita boundary demonstrates the practical application of this design, showing that administrative borders can reveal persistent causal effects when these identifying assumptions are satisfied. Because the ULEZ expansion is defined by an administrative boundary, a geographic regression discontinuity design is available in principle. However, the limited number of boroughs adjacent to the boundary and the presence of broader administrative differences make these identifying assumptions considerably more demanding than those underpinning the synthetic control approach. Section 6 assesses this design for feasibility rather than adopting it as a robustness check.

Taken together, these methodological contributions establish the principles that guide the empirical strategy adopted in this dissertation. The analysis requires a design that constructs an explicit counterfactual rather than assuming parallel trends, compares treated units only with areas plausibly unaffected by the relevant policy announcement, and supports credible inference despite a limited number of treated units. Synthetic control therefore serves as the primary research design, implemented using never-treated surrounding districts alongside not-yet-treated London boroughs as donor units, with placebo-based inference. Two-way fixed effects difference-in-differences, event-study and interrupted time-series specifications provide complementary robustness checks, while a geographic regression discontinuity design is assessed for feasibility but cannot be credibly identified at borough level.

4 Data and Study Design

4.1 Data Sources and Construction

The outcome of interest is the share of licensed cars registered within each local authority that are classified as Ultra Low Emission Vehicles (ULEVs). Vehicle registration data are obtained from the UK Driver and Vehicle Licensing Agency (DVLA) quarterly licensing statistics (Tables VEH0132 and VEH9901). The ULEV classification follows Department for Transport convention, covering vehicles emitting below the prevailing carbon dioxide threshold, principally battery electric and plug-in hybrid cars. The numerator is the number of licensed ULEVs registered within each local authority, while the denominator is the total number of licensed cars registered in the same area. The resulting measure captures the proportion of the local vehicle fleet classified as ULEV. Importantly, this is a stock rather than a flow measure: it reflects the composition of the existing fleet rather than new vehicle registrations. Consequently, any behavioural response to ULEZ announcements is expected to emerge gradually as households replace or retire vehicles over time rather than instantaneously.

As discussed in Section 2, ULEV ownership is not synonymous with ULEZ compliance. Petrol vehicles meeting Euro 4 standards and diesel vehicles meeting Euro 6 standards are equally exempt from the ULEZ charge despite not being classified as ULEVs. The ULEV share is adopted because it captures the form of adjustment that exceeds the compliance threshold rather than merely meeting it. A household that replaces a non-compliant vehicle with a compliant diesel satisfies the announced charge; one that purchases a ULEV commits to a lower-emission vehicle beyond what the regulation requires, a decision more plausibly reflecting expectations about the future direction of regulation than minimal adjustment to a known cost. This is the behavioural margin the theoretical framework predicts. It also has an important implication for interpretation: the estimated effects should be regarded as a lower bound on the total compliance response, since households that replaced non-compliant vehicles with compliant conventional petrol or diesel vehicles are not observed in the outcome measure.

The empirical analysis uses a balanced quarterly panel comprising 52 local authorities observed from 2011 Q4 to 2024 Q4, yielding 2,756 observations. The panel consists of the 33 Greater London local authorities (comprising the 32 London boroughs and the City of London) and 19 surrounding district local authorities. The surrounding districts are selected using a transparent geographic rule. Greater London's boundary is defined as the union of the 33 borough geometries, and all district authorities with any part of their boundary lying within two kilometres of Greater London are included in the study panel. This criterion identifies districts immediately adjoining London while avoiding progressively more distant authorities that are less comparable in terms of travel patterns, housing markets, and vehicle ownership. These surrounding districts were never themselves subject to the ULEZ charge, although many of their residents commute into London, making them a plausible source of comparison units throughout the analysis.

In addition to the vehicle licensing data, the panel includes several socio-economic and transport characteristics used in robustness and heterogeneity analyses. These comprise the Index of Multiple Deprivation (IMD), Public Transport Accessibility Levels (PTAL), median house prices, car ownership, population density, and household income. These variables are not used to construct the synthetic control weights themselves, which are based solely on the pre-treatment outcome trajectory, but instead provide additional evidence on the robustness and heterogeneity of the estimated effects.

4.2 Treatment Timing and the Dating Rule

The empirical analysis dates each ULEZ expansion to the point at which its implementation became a credible and firm policy commitment rather than to the date on which charging commenced. This approach follows directly from the theoretical framework developed in Section 3. If forward-looking agents respond to credible announcements of future regulatory costs, then behavioural adjustment should begin once implementation becomes certain, not when the policy is implemented. Accordingly, the treatment date for each expansion is defined as the earliest point at which the public could reasonably regard the expansion as a committed policy rather than a proposal subject to revision.

Table 1: Treatment dates by expansion wave

ULEZ Expansion	Treatment Date	Rationale
Central London (2019)	2017 Q2 (4 April 2017)	The central ULEZ had already been confirmed under the previous Mayor in March 2015 for implementation in 2020. The April 2017 announcement brought the implementation date forward to April 2019, altering the timing of an already-committed policy rather than creating a new commitment.
Inner London (2021)	2018 Q2 (8 June 2018)	The same April 2017 announcement set out an intended extension to the North and South Circular roads, but implementation remained subject to statutory consultation. The policy became a firm commitment only when the expansion was formally confirmed following that consultation.
Outer London (2023)	2022 Q4 (25 November 2022)	The March 2022 consultation proposed a London-wide expansion, with implementation contingent on the statutory consultation process. The confirmed decision of 25 November 2022 represents the point at which the expansion became a credible commitment.

The distinction between the 2019 and 2021 expansions is particularly important because both are referenced in the same April 2017 announcement, yet that announcement conveyed different information for each policy. For the 2019 central London ULEZ, the underlying commitment already existed: the policy had been confirmed in 2015 and the April 2017 announcement simply accelerated its implementation from 2020 to 2019. For the 2021 inner London expansion, by contrast, the same document announced an intended future extension that remained subject to statutory consultation before implementation could be confirmed. Although both policies appeared in the same publication, they occupied different stages of the policy process. The informational content of the announcement therefore differed, with one confirming revised timing for an existing commitment and the other signalling an intention that had not yet acquired the same level of institutional commitment.

To assess the sensitivity of the results to this dating rule, alternative announcement dates are examined as robustness checks. For the 2019 expansion, an alternative specification dates treatment to the November 2017 confirmation rather than the April 2017 acceleration announcement. Conversely, for the 2021 expansion, an alternative specification dates treatment to the initial April 2017 proposal rather than the June 2018 confirmation. These robustness exercises apply symmetric alternative dating rules across the two policy waves and therefore address the possibility that the estimated anticipatory effects are sensitive to the precise definition of treatment timing.

The resulting anticipation windows differ across the three expansions. Approximately two years separate the April 2017 announcement from the introduction of the central London ULEZ in April 2019, while the interval between the June 2018 confirmation and the October 2021 inner London expansion is approximately three years and four months. The outer London expansion provides a substantially shorter anticipation period of around nine months between confirmation in November 2022 and implementation in August 2023. These differences reflect the institutional timetable of each expansion rather than arbitrary modelling choices, and are reflected in the wave-specific estimation windows described below.

4.3 Donor Pool Construction

The validity of the synthetic control method depends not only on the quality of the match between treated and comparison units, but also on the composition of the donor pool from which the synthetic control is constructed. As discussed in Section 3, Abadie (2021) argues that donor units should be plausibly unaffected by the intervention under study, while excessively large or heterogeneous donor pools risk weakening the credibility of the constructed counterfactual. Consequently, donor pool construction is treated as a substantive component of the research design rather than a purely technical decision.

Because this dissertation studies anticipatory responses to policy announcements rather than responses to policy implementation, treatment status is defined using the announcement dates established in Section 4.2. A local authority is therefore considered untreated only until the announcement that credibly commits it to a future ULEZ expansion. This operational definition follows directly from the theoretical framework developed in Section 3 and is consistent with the emphasis of Goodman-Bacon (2021) and Callaway and Sant'Anna (2021) on using never-treated or not-yet-treated comparison groups in settings with staggered policy adoption. Under an announcement-based design, "not yet treated" refers to units that have not yet been exposed to a credible policy announcement rather than those that have not reached the implementation date.

The resulting donor pools differ across the three ULEZ expansions.

Table 2: Donor pool composition by expansion wave

Expansion	Donor Pool Size	Composition
2019	38	19 surrounding districts + 19 outer London boroughs
2021	38	19 surrounding districts + 19 outer London boroughs
2023	19	19 surrounding districts

The principal consequence of this definition concerns the 2019 analysis. Although the 2021 expansion was not implemented until October 2021, six inner London boroughs — Hammersmith and Fulham, Haringey, Kensington and Chelsea, Lewisham, Newham, and Waltham Forest — fell within the extension boundary set out in the mayoral announcement of 4 April 2017. From the perspective of an announcement-based design, these boroughs therefore became treated at the same point as the 2019 expansion, even though charging did not commence for several years. Including them as donor units would require comparison areas to have already received the very policy signal whose behavioural consequences the analysis seeks to identify. They are therefore excluded from the donor pool for the 2019 analysis.

By contrast, the outer London boroughs remain eligible donor units for both the 2019 and 2021 analyses. Although the 4 April 2017 announcement set out changes affecting heavy vehicles in outer London from 2020, these measures did not apply to private cars, which constitute the outcome examined in this paper. The relevant announcement for private vehicle owners occurred only with the confirmation of a London-wide ULEZ expansion in November 2022. Consequently, outer London boroughs satisfy the announcement-based definition of untreated for the earlier policy waves.

For the 2023 expansion, however, the donor pool is necessarily restricted to the 19 surrounding districts. By the time the London-wide expansion was confirmed in November 2022, every London borough had either already entered the ULEZ or had been credibly committed to joining it. No London local authority therefore satisfied the announcement-based definition of untreated. The districts-only donor pool is thus imposed by the institutional history of the policy rather than selected as a matter of methodological preference, though the reduction in pool size has direct consequences for inference, discussed in Section 4.6.

The inclusion of the 19 surrounding district authorities represents an important feature of the research design. These districts share labour markets, commuter flows, housing markets, and vehicle markets with outer London while never themselves being subject to the ULEZ charge. They therefore provide comparison units that are geographically and economically similar to the treated boroughs without experiencing the direct policy intervention. This contrasts with studies of nationwide interventions, such as Mills and Rüttenauer's (2022) analysis of COVID-19 certification, where all units are exposed simultaneously and no genuinely untreated geographic comparison group exists. The surrounding districts consequently strengthen the credibility of the constructed counterfactual across all three ULEZ expansions.

A geographic regression discontinuity design was also considered as a complementary identification strategy, because the ULEZ boundary constitutes a genuine administrative discontinuity. However, its feasibility depends on the availability of sufficient comparable units close to the boundary, and it is therefore evaluated separately as a feasibility assessment in Section 6. The empirical consequences of the donor pool construction adopted here are likewise examined in Section 6, while the question of whether surrounding districts themselves respond to ULEZ announcements is addressed in the following subsection.

4.4 What the Design Identifies

The donor pool construction described in Section 4.3 is based on the methodological requirement that comparison units should be plausibly unaffected by the treatment under study. In the present setting, however, it is important to distinguish between policy coverage and behavioural exposure. Although the 19 surrounding districts were never themselves subject to the ULEZ charge, many of their residents commute into London and may therefore have responded to announcements of future ULEZ expansions. Consistent with this possibility, the DiD-ITS analysis presented in Section 6 identifies a positive level shift in ULEV share of approximately 0.022 among the surrounding districts coinciding with the November 2022 announcement, a shift no smaller than that observed across outer London itself.

This common movement does not invalidate the synthetic control design because the estimator is constructed to remove changes shared by the treated unit and its donor pool. Conceptually, the observed change in any treated borough may be decomposed into a component common across the wider region and a borough-specific component attributable to differential behavioural responses. By matching each treated borough to a weighted combination of donor units with similar pre-treatment trajectories, synthetic control differences out the common regional component by construction. If the treated borough and its synthetic counterpart experience the same acceleration following an announcement, the estimated treatment effect is zero, irrespective of whether both series increased in absolute terms. The aggregate null effect estimated for the 2023 expansion is therefore entirely consistent with widespread change in ULEV adoption across both London and the surrounding districts, provided that the magnitude of that change was similar in treatment and comparison areas.

This identifying strategy has an important implication for interpretation. The estimated effects should not be interpreted as measuring total anticipatory compliance generated by ULEZ announcements. Instead, they identify differential anticipatory compliance, that is, behavioural responses that exceed those observed among otherwise comparable comparison areas. Uniform changes occurring across the wider London travel-to-work region are intentionally removed by the estimator and therefore do not appear in the estimated treatment effects. This scope condition applies throughout the dissertation and should be borne in mind when interpreting the empirical results. The mechanisms through which such common movement may have arisen are considered in Section 8.

4.5 Estimation

The synthetic control estimator is constructed using only information observed before the relevant policy announcement. The matching specification employs two predictors derived from the outcome variable itself: the mean pre-treatment ULEV share over the optimisation window and the ULEV share in the final quarter immediately preceding the announcement. No external covariates enter the matching procedure. Instead, the optimisation relies on the historical evolution of the outcome variable itself.

The matching process extends beyond these two predictors alone. The optimisation algorithm selects donor weights that minimise prediction error across every quarter within the pre-treatment window, subject to the synthetic control constraints. The predictors anchor the optimisation by defining the moments to be matched, while the quarterly outcome trajectory determines the weights that best reproduce the observed evolution of the treated borough. The synthetic control is therefore fitted to the entire pre-announcement path rather than to a small set of summary statistics. This approach follows the guidance of Abadie (2021), who argues that the pre-treatment outcome path captures the combined influence of both observed and unobserved determinants of the outcome, reducing the need to include additional covariates directly in the matching procedure.

Different optimisation windows are used for each expansion to reflect their respective announcement dates while maintaining comparability where possible. For the 2019 expansion, the optimisation window spans 2014 Q1 to 2017 Q2, comprising 14 pre-treatment quarters. The 2021 expansion uses an equally long window from 2015 Q1 to 2018 Q2, again covering 14 quarters. Maintaining identical window lengths for these two waves facilitates comparison of the resulting synthetic controls despite their different announcement dates. The 2023 expansion uses the period from 2019 Q1 to 2022 Q4, providing 16 pre-treatment quarters before the November 2022 announcement. The window begins in 2019 Q1, after the central London zone came into force. It necessarily spans the October 2021 inner London implementation, but neither the treated outer boroughs nor the surrounding districts fell within that zone, so any resulting influence is common to both groups and is differenced out by the estimator.

For the 2019 and 2021 analyses, the combination of 38 donor units and 14 pre-treatment observations frequently produces extremely small pre-treatment mean squared prediction errors (MSPEs). Such close fits are expected when the donor pool is sufficiently rich to reproduce the treated unit's historical trajectory. Close pre-treatment fit does not by itself invalidate the results. Because the same optimisation procedure is applied to every placebo unit, any tendency towards overfitting affects both the treated borough and the placebo distribution equally. The validity of the subsequent inference therefore depends not on the absolute size of the pre-treatment MSPE but on whether the treated borough exhibits a post-treatment divergence that is unusually large relative to those observed in the placebo exercises.

4.6 Inference

Inference is based on the placebo framework developed by Abadie, Diamond, and Hainmueller (2010, 2015) and discussed in Section 3. Rather than relying on conventional sampling theory, synthetic control evaluates the estimated treatment effect by comparing it with the distribution of placebo effects obtained by iteratively assigning treatment to each donor unit. The estimated effect for the treated borough is therefore interpreted relative to the effects observed when the same estimation procedure is applied to units that were not exposed to the policy announcement. The corresponding p-value is given by the rank of the treated unit within the placebo distribution, ordered by the magnitude of the estimated effect, divided by the total number of treated and placebo units. A rank of one therefore indicates the largest divergence in the sample, and the resulting p-value provides a non-parametric measure of whether the observed effect is unusually large under the null hypothesis of no treatment effect.

Because placebo inference depends on the number of available comparison units, the attainable p-values are bounded by the size of the donor pool. For the 2019 and 2021 analyses, the donor pool contains 38 donor units, yielding 39 total units and a minimum attainable p-value of 0.026. For the 2023 analysis, the districts-only donor pool contains 19 donor units, giving 20 total units and a minimum attainable p-value of 0.05. Consequently, boroughs such as Hillingdon and Sutton, which achieve p-values of exactly 0.05, are ranked first in the placebo distribution and cannot attain a smaller p-value regardless of the magnitude of their estimated effects. P-values should therefore be interpreted alongside the size and persistence of the estimated treatment effects rather than in isolation, and direct comparison of p-values across the 2023 and earlier policy waves should be made with appropriate caution.

For this reason, inference also considers the ratio of post-treatment to pre-treatment mean squared prediction error (MSPE), as proposed by Abadie, Diamond and Hainmueller (2015). The MSPE ratio evaluates the magnitude of the estimated treatment effect relative to the quality of the pre-treatment fit, providing a scale-independent measure of divergence between the treated borough and its synthetic counterpart. This statistic is particularly informative when pre-treatment fit differs across units or when the minimum attainable p-value is constrained by the size of the donor pool. The City of London illustrates why absolute gaps can mislead. Its estimated anticipation gap is the largest among the 2019 treated boroughs, yet its MSPE ratio is only 3.7, because the synthetic control never reproduced its pre-treatment trajectory closely. A large post-treatment divergence carries little evidential weight when the pre-treatment match was poor, and the borough accordingly ranks near the bottom of the placebo distribution. Considering both the placebo rank and the MSPE ratio therefore provides a more complete assessment of statistical evidence than either measure alone.

The credibility of the findings is further assessed through a series of robustness analyses motivated by the methodological considerations discussed in Section 3. These include two-way fixed effects difference-in-differences models using never-treated surrounding districts as comparison units, event-study specifications examining the dynamics of treatment effects, interrupted time-series and difference-in-differences interrupted time-series analyses, and alternative announcement dates for the 2019 and 2021 expansions. The feasibility of a geographic regression discontinuity design is also evaluated. Together, these complementary analyses assess the sensitivity of the principal findings to alternative identification strategies and modelling assumptions while retaining synthetic control as the primary empirical framework. Heterogeneity in the estimated responses, disaggregated by vehicle keepership and fuel type, is examined separately in Section 7.

5 Results

This section reports the synthetic control estimates for the three Ultra Low Emission Zone (ULEZ) expansion announcements, using the empirical specifications described in Section 4. Results are presented separately for the 2019 central London expansion, the 2021 inner London expansion, and the 2023 London-wide expansion. Table 3 provides a consolidated summary of the principal estimates for each expansion, while full results for all treated boroughs are reported in the appendix. Following the framework set out in Section 4.6, statistical significance is assessed jointly using each treated unit's placebo rank and its post-/pre-treatment MSPE ratio, rather than the estimated anticipation gap in isolation.

Table 3: Synthetic control estimates by expansion wave. Boroughs with $p \leq 0.20$ are shown; full results appear in Appendix B.

Expansion	Borough	Anticipation gap	MSPE ratio	Rank	p-value
2019	Tower Hamlets	0.002	752.563	3	0.077
2021	Hammersmith and Fulham	0.015	3304.946	3	0.077
2021	Newham	0.004	2212.112	4	0.103
2023	Hillingdon	0.008	197.188	1	0.050
2023	Sutton	0.002	222.307	1	0.050
2023	Barking and Dagenham	0.002	78.525	3	0.150
2023	Greenwich	-0.005	52.984	3	0.150
2023	Harrow	0.004	80.228	3	0.150
2023	Richmond upon Thames	-0.001	62.983	3	0.150
2023	Wandsworth	-0.027	51.467	3	0.150
2023	Croydon	0.001	38.919	4	0.200
2023	Kingston upon Thames	0.002	25.554	4	0.200

5.1 The 2019 Central Expansion

Among the eight boroughs included in the 2019 expansion, Tower Hamlets provides the strongest evidence of an anticipatory response. It records the third-highest post-/pre-treatment MSPE ratio within its placebo distribution (753), corresponding to a placebo rank of 3 out of 39 and a permutation p-value of 0.077 (Table 3). Although this represents the strongest treated result in the wave, it does not reach conventional levels of statistical significance.

Figure 2 plots the observed and synthetic trajectories for Tower Hamlets. The treated borough closely follows its synthetic counterpart throughout the pre-announcement period, indicating good pre-treatment fit. Following the announcement in 2017 Q2, a positive divergence emerges and persists over the anticipation period. The estimated anticipation gap is 0.0025, rising to 0.0078 over the full post-announcement period, which extends beyond implementation in April 2019.

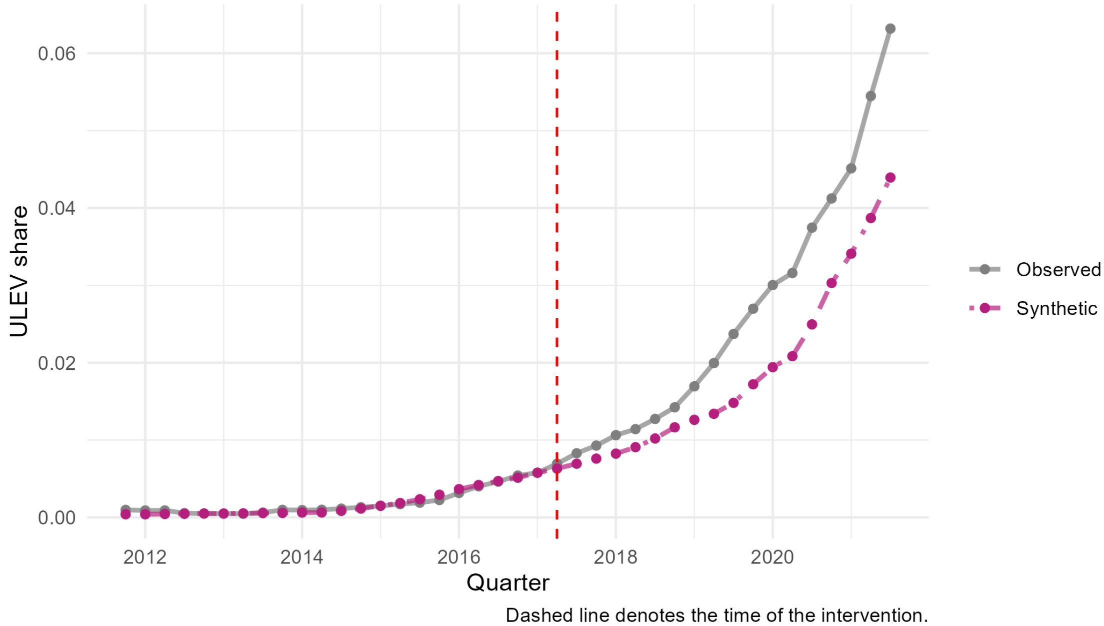

Figure 2: Tower Hamlets tracks its synthetic control closely throughout the pre-announcement period, after which a modest positive gap opens and persists beyond implementation.

The result remains marginal because two donor units exhibit larger MSPE ratios than Tower Hamlets. Windsor and Maidenhead records the highest ratio (1123), followed by Merton (1106), leaving Tower Hamlets ranked third within the placebo distribution. Under the permutation framework described in Section 4.6, this implies that the observed divergence cannot be distinguished from the variation observed among units that received no announcement at the 5% significance level.

The contrast with the City of London illustrates why estimated gap magnitude alone is not sufficient evidence of an anticipatory effect. The City of London records the largest estimated anticipation gap in the wave (0.0212), more than eight times that of Tower Hamlets, yet its MSPE ratio is only 3.7, ranking 37th among the placebo units. This reflects poor pre-treatment fit, so the large estimated gap provides little evidential support for an anticipatory response.

The remaining treated boroughs — Islington, Camden, Hackney, Westminster, Southwark, and Lambeth — show no evidence of anticipatory effects under the primary specification, with permutation p-values ranging from 0.615 to 0.949. Complete results for all treated boroughs are reported in Appendix B.

5.2 The 2021 Inner Expansion

No treated borough attains conventional levels of statistical significance under the primary specification for the 2021 inner London expansion. Across the six boroughs entering the ULEZ in this wave, the synthetic control estimates provide no evidence of anticipatory changes in ULEV share that can be distinguished from placebo variation (Table 3).

Hammersmith and Fulham provides the closest case to statistical significance. It records the third-highest MSPE ratio within its placebo distribution, corresponding to a placebo rank of 3 and a permutation p-value of 0.077. The estimated anticipation gap is 0.0148, increasing to an average post-announcement gap of 0.0287. Despite this divergence, the result remains below conventional thresholds for statistical significance.

The remaining treated boroughs provide weaker evidence. Newham records a permutation p-value of 0.103, while Waltham Forest records 0.256, Lewisham 0.462, and Haringey 0.692. Kensington and Chelsea exhibits a negative estimated anticipation gap relative to its synthetic control.

Taken together, the results indicate an absence of statistically significant anticipatory responses across all six boroughs included in the 2021 expansion under the primary specification. This finding is observed despite the 2021 expansion having the longest anticipation period of the three policy waves. In contrast to the marginal evidence observed for the 2019 expansion and the significant results identified in two boroughs during the 2023 expansion, the 2021 estimates remain null at conventional thresholds under the primary analysis.

5.3 The 2023 London-wide Expansion

Two boroughs exhibit statistically significant anticipatory responses under the primary specification: Hillingdon and Sutton. Both record the highest post-/pre-treatment MSPE ratio within their respective placebo distributions, corresponding to a placebo rank of 1 out of 20 and a permutation p-value of 0.05 (Table 3). As discussed in Section 4.6, the 2023 analysis uses a donor pool of 19 untreated districts, meaning that 0.05 is the smallest attainable permutation p-value. Consequently, a rank of 1 represents the strongest evidence that the synthetic control design can produce under this specification rather than a borderline result.

Figure 3 shows the observed and synthetic trajectories for Hillingdon and Sutton. In both boroughs, the observed series closely tracks its synthetic counterpart throughout the pre-announcement period before diverging after the announcement in 2022 Q4. The estimated anticipation gap is 0.0081 for Hillingdon and 0.0022 for Sutton, rising to 0.0151 and 0.0048 respectively over the full post-announcement period. Sutton records the higher MSPE ratio (222 against 197) despite the smaller estimated gap, reflecting a closer pre-treatment match.

Beyond these two cases, the results provide little evidence of systematic anticipatory effects across the remaining outer London boroughs. Barking and Dagenham and Harrow each record permutation p-values of 0.15, while Croydon records 0.20. Nine of the nineteen treated boroughs exhibit negative estimated anticipation gaps relative to their synthetic controls, the largest being Wandsworth at -0.027. The distribution of estimated effects across outer London is therefore centred close to zero rather than uniformly positive, with complete results reported in the appendix.

As noted in Section 4.4, the estimated effects represent divergences relative to the untreated comparison districts. Any response common to both treated boroughs and donor units is therefore removed by construction and would not appear in these estimates.

Across the three expansion waves, the synthetic control estimates show differing patterns of

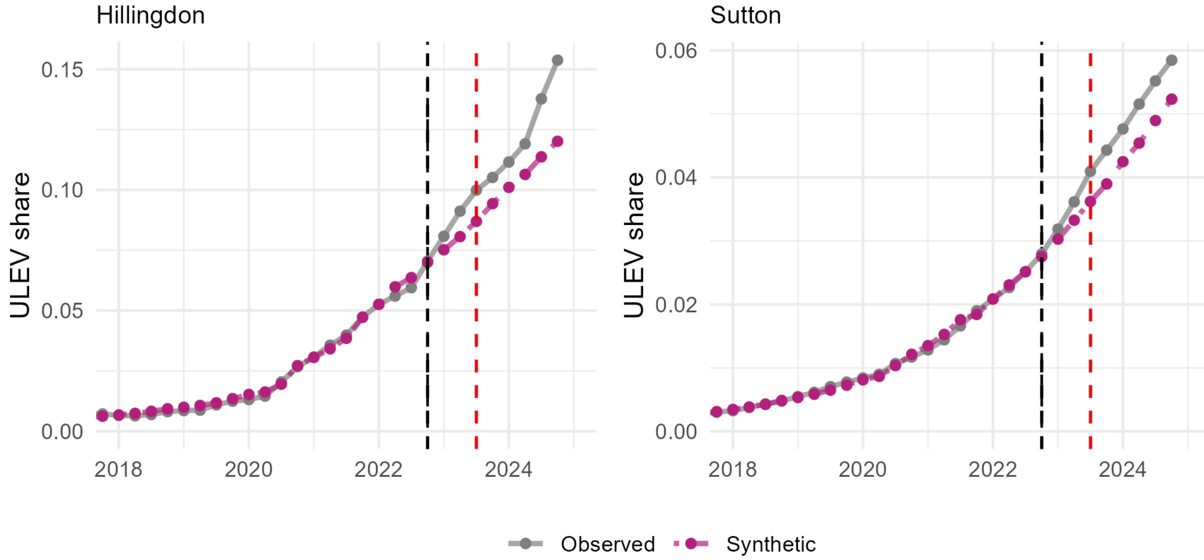

Figure 3: Hillingdon and Sutton both track their synthetic controls closely before the November 2022 announcement, after which positive gaps open and persist through implementation.

anticipatory response. The 2019 central expansion provides the strongest evidence, although the Tower Hamlets estimate remains marginal under the placebo distribution. The 2021 inner expansion produces no statistically significant effects across the treated boroughs, while the 2023 London-wide expansion identifies significant results for Hillingdon and Sutton alongside an aggregate null across the remaining boroughs. Estimated effects are modest throughout, with the largest anticipation gap among the significant cases below one percentage point of the licensed fleet. This pattern is consistent with the prediction in Section 3.1 that responses would be partial and heterogeneous rather than uniform. Section 6 assesses the robustness of these estimates, while Section 8 returns to their interpretation.

6 Robustness Checks

7 Heterogeneity Analysis

8 Discussion and Conclusion